ANY THICKENING OF AN AFFINE SCHEME IS AFFINE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 156-165 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 167-206, the complete author proof (38 body lines).

Editorial note. Navigation stubs: 6 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Pool card span [167,170] disagreed with the recomputed span and was ruled a line-number defect (Chief ruling 2026-10-13, LANE_BRIEF §16.5/§16.6: real spans are computed from the \label, the statement union the whole proof environments); card follows the recomputed span.

Editorial note. The statement's reference block contains the chapter's only \cite{EGA1} in this span; satisfied by the card-local bibliography.tex transcribed from the snapshot my.bib entry EGA1 (Institute des Hautes Etudes Scientifiques, Publications Mathematiques 4, 1960).

Editorial note. Chapter opening: first lemma of the thickening series (definition-thickening stub carried with it).

Editorial note. Pool-card span [167,170] covered only the pointer proof; the collected proof span 167-206 adds the substantive finite-order-thickening proof per the Lead ruling of 2026-10-13 (real spans recomputed from the \label). The statement's reference block \cite{EGA1} is satisfied by a card-local bibliography.tex transcribed from the snapshot my.bib EGA1 entry; bibliography_backend none (thebibliography environment directly).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Limits, proposition limits-proposition-affine; Cohomology of Schemes, lemmas coherent-lemma-quasi-compact-h1-zero-covering, coherent-lemma-relative-affine-cohomology, coherent-lemma-quasi-coherent-affine-cohomology-zero; Morphisms, lemma morphisms-lemma-i-star-equivalence; EGA1, Proposition 5.1.9 (cited in the statement's reference block).

Source reference. Original lemma 01XB: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 206–206. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01XF: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 206–206. Cited by the retained statement or proof

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but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 089W: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 206–206. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 04EX: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 36–63. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 05YU: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 206–206. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01QY: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 206–206. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Affineness is insensitive to thickenings The case of a finite order thickening is* [1, Proposition 5.1.9]. *Any thickening of an affine scheme is affine.*

Proof. This is a special case of Limits, Proposition 05YU. □

Proof for a finite order thickening. Suppose that $X \subset X'$ is a finite order thickening with X affine. Then we may use Serre's criterion to prove X' is affine. More precisely, we will use Cohomology of Schemes, Lemma 01XF. Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_{X'}$-module. It suffices to show that $H^1(X', \mathcal{F}) = 0$. Denote $i : X \to X'$ the given closed immersion and denote $\mathcal{I} = \mathrm{Ker}(i^\sharp : \mathcal{O}_{X'} \to i_*\mathcal{O}_X)$. By our discussion of finite order thickenings (following Definition 04EX) there exists an $n \geq 0$ and a filtration

$$0 = \mathcal{F}_{n+1} \subset \mathcal{F}_n \subset \mathcal{F}_{n-1} \subset \ldots \subset \mathcal{F}_0 = \mathcal{F}$$

by quasi-coherent submodules such that $\mathcal{F}_a/\mathcal{F}_{a+1}$ is annihilated by $\mathcal{I}$. Namely, we can take $\mathcal{F}_a = \mathcal{I}^a\mathcal{F}$. Then $\mathcal{F}_a/\mathcal{F}_{a+1} = i_*\mathcal{G}_a$ for some quasi-coherent $\mathcal{O}_X$-module $\mathcal{G}_a$, see Morphisms, Lemma 01QY. We obtain

$$H^1(X', \mathcal{F}_a/\mathcal{F}_{a+1}) = H^1(X', i_*\mathcal{G}_a) = H^1(X, \mathcal{G}_a) = 0$$

The second equality comes from Cohomology of Schemes, Lemma 089W and the last equality from Cohomology of Schemes, Lemma 01XB. Thus $\mathcal{F}$ has a finite filtration whose successive quotients have vanishing first cohomology and it follows by a simple induction argument that $H^1(X', \mathcal{F}) = 0$. □

References

[1] Grothendieck, Alexander and Dieudonné, Jean. *Éléments de géométrie algébrique I.* Institute des Hautes Études Scientifiques. Publications Mathématiques, Vol. 4, 1960.